

We began by explaining that Earth has two major ice sheets, one in Antarctica and one in Greenland. Even small changes in these huge bodies of ice can change the volume of the oceans and how often coastal floods occur. Using a slide labeled “Sea level is rising,” he showed measurements of sea level over the last few decades. Then he presented a “Projected sea level” graph from the IPCC, which showed how sea level could rise under different emissions scenarios. He pointed out that a lot of the uncertainty in these projections comes from not knowing exactly how the ice sheets will respond.

He then described why ice sheets are so hard to study. Antarctica is thousands of kilometers across, with ice thicker than skyscrapers, and the continent beneath is mostly below sea level and buried under ice. Snow piles up, fractures, and slowly deforms, spreading outward like a very slow flowing liquid. The ice has to move toward the edges of the continent; if it did not, it would just keep building up. Around the somewhat warmer coasts, that flowing ice reaches the edges and starts to melt. We saw colorful maps of Antarctic surface elevation and complex ice flow patterns, then explained how water controls much of this motion. Meltwater can drain through the ice to the bed, where it changes the friction between ice and rock. The pressure of this water at the base can lubricate the ice and either speed it up or slow it down. He finished with slides titled “Water moves across the surface of ice shelves,” which showed satellite images of meltwater streams and ponds on floating ice shelves around Antarctica. These images reinforced his main point: the paths that water takes through and across ice sheets strongly affect how quickly they change and how much they raise global sea level.

